DEEP-DIVE ARCHITECTURAL & LOGIC AUDIT #06A: PILLS & ALCHEMY SYSTEM (PART 1 — BASIC WELFARE, VITAL RECOVERY, INJURY-MENDING, DETOX & MIND-CALMING PILLS)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` / `SIM.PATCH_VERSION = '2.6.0'` / `HC.FINGERPRINT = '155:2f9c9dd7'`)
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini** Models)
- **Master Roadmap for Series `06A–06F` (Complete Granular Pill & Alchemy Ecosystem Audit)**:
  - **`06A` (This Document)**: Group 1 — **Basic Welfare, Vital Recovery, Injury-Mending, Detox & Mind-Calming Pills (`JD.CP.BASIC` 8 Pills, `REC140_ALC_QI_DEW_1 / JADE_MARROW_2 / SNOW_SOUL_5`, & `ITEM190_PILL_001..005, 008`)**
  - `06B`: Group 2 — Cultivation-Acceleration, Meridian/Marrow/Bloodline-Tempering, Insight, Fish-Alchemy & Longevity Elixirs
  - `06C`: Group 3 — Realm Breakthrough Pills (`P:BT_*`), Bottleneck Treasures (`BT.TREASURES`), Pill Resistance (`btTaken`) & Wrong-Path Backlash
  - `06D`: Group 4 — Pre-Fight Combat Buff Pills (`P:CB_*`), Tactical/Stealth/Control-Resilience Pills, Beast-Bond Pills & Forbidden Blood-Sacrifice Pills
  - `06E`: Group 5 — Multi-Phase Specialty Elixirs (`PILL_V234_001..048`), Dynamic Elemental Fusion Pills (`PILL_V235_FUSION_*`) & Experimental Formulas (`content.js`)
  - `06F`: Part 6 — Grand Finale: Holistic Overview of the Entire Pill, Toxicity & Alchemy Ecosystem (`4` Consumption Pathways, `applyQualityPill` Bug, Dual-ID Stock Split & Master Roadmap)

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0. Hard Architectural Invariants for AI Implementers

1. **Immutable `human_core.js`**: Never modify `GAME_SOURCE/js/human_core.js` (`HC.FINGERPRINT === '155:2f9c9dd7'`).
2. **Closed-Loop Stock & Vow Respect**: Autonomous welfare distribution and emergency pill consumption must **never** force-feed pills to ascetics upholding

`VOW_ABSTAIN_PILLS` / `VEGETARIAN_PACIFIST` (`daoOath`) or `VOW_NO_MEDICINE` (`voluntaryVows`), and must never mint nonexistent ghost pill keys (`'P:HEAL'`).

3. **Combat Anti-Heal Integrity**: Mid-battle pill healing (`CPE.fightUse`) must respect active anti-heal statuses (`healBlock`, `BLOOD_HALT`, `stygianFreeze`).

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1. `[PILL-REC-01]` `100×` Unit-Scale Mismatch on `qiRestore` (`economy.js:322` vs `data_cp.js:11` vs `cpeco.js:396`) & Monkey-Patch Ordering Trap!

Trace `qiRestore` across the three files that define and consume it:

1. **`data/data_cp.js:11–12` (Fractional `0..1` Scale)**:
   - `BASIC_QI`: `{ qiRestore: 0.5, toxicity: 1 }` (`50%` Max Qi)
   - `BASIC_QI2`: `{ qiRestore: 0.9, toxicity: 2 }` (`90%` Max Qi)
2. **`js/economy.js:321–322` (Percentage `0..100` Scale on Item-Defined Pills!)**:

```
JavaScript
if (it.healPct && !e.healHealth) e.healHealth = Math.round(it.healPct * 100); // 0.42 -> 42
if (it.qiPct && !e.qiRestore) e.qiRestore = Math.round(it.qiPct * 100);       // 0.45 -> 45 (NOT 0.45!!)
```

3. **`js/cpeco.js:395–397` (Multiplies `qiMax * e.qiRestore` Without Dividing `> 1` Values by `100`!)**:

```
JavaScript
if (e.qiRestore && npc.cult) {
  const q = CULT.qiMax(npc);
  s.qi = Math.min(q, (s.qi == null ? 0 : s.qi) + q * e.qiRestore);
}
```

- **Three Bugs Caused by This Unit Split**:
  1. **`100×` Over-Scaling on Item Pills**: Any pill with `it.qiPct = 0.25` (`25%` Qi) converts to `e.qiRestore = 25`, restoring **`2,500%` Max Qi** (`q * 25`, instantly capping `100%` Qi regardless of `0.70` poor quality or Qi-suppression!).
  2. **Missing Base Handler in `economy.js:338–355` & Broken `stats.js:658` Capture**: `economy.js`'s base `ECO.applyPill` **completely omits `qiRestore`, `energyRestore`, and `restRestore`**, relying on `cpeco.js:395`'s

wrapper—yet `stats.js:658` (`const oApplyPill = ECO.applyPill`) loads *before* `cpeco.js` in `index.html`, capturing the unwrapped version!

3. **Undocumented `* 1.2` Heal Inflation (`economy.js:343`)**: `s.hp + CULT.hpMax(npc) * e.healHealth / 100 * 1.2` silently inflates every heal pill by `+20%` (`P:BASIC_HP2` `70` heals `84%`, `q = 1.3` heals `109%`), outclassing higher-tier authored heal pills!

- **Exact Fix**: Normalize `const qiFrac = e.qiRestore > 1.5 ? e.qiRestore / 100 : e.qiRestore`, move `qiRestore` / `energyRestore` / `restRestore` directly into base `ECO.applyPill` in `economy.js` (removing the duplicate wrapper in `cpeco.js:395`), and calibrate `healHealth / 100` honestly!

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2. `[PILL-REC-02]` Autopilot Force-Feeding Shatters Ascetic Vows (`VOW_ABSTAIN_PILLS` & `VOW_NO_MEDICINE`) Across `sim.js` & `cpeco.js`!

- **Locations**: `js/sim.js` (`SIM.hourly`), `js/cpeco.js:250–315` (`CPE.pickPurchase`), `419–435` (`CPE.autoUse` & `CPE.fightUse`), `440–518` (`A.personalBrew`, `A.personalPillUse`, `CPE.welfareDaily`)
- **Verified Tragic Cascade**:
  1. **Zero Vow Checks in `cpeco.js` or `sim.js`**: Neither `SIM.hourly` nor **any** function in `cpeco.js` checks `n.daoOath` (`VOW_ABSTAIN_PILLS` / `VEGETARIAN_PACIFIST`) or `n.voluntaryVows.VOW_NO_MEDICINE`!
  2. The moment an ascetic monk takes `32%` chip damage:
     - `CPE.welfareDaily` hands them `P:BASIC_HP` (or `SIM.hourly` force-feeds a sect basic pill).
     - `CPE.autoUse` / `A.personalPillUse` (`score = 2.45`!) immediately swallows it.
     - `ECO.applyPill` (`economy.js:339`) triggers `CULT.breakOathVow` $\to$ **shattering their vow, inflicting `+35 heartDemon` and `'สัตย์สาบานพังทลาย'` spiritual injury**!
     - On the very next tick, seeing `heartDemon >= 35` and `n.inj.length > 0`, `CPE.pickPurchase` and `CPE.autoUse` **force them to buy and swallow `P:BASIC_WOUND` and `P:BASIC_CALM` as well**!
  3. **Asymmetric Vow Enforcement (`economy.js:339` vs `379`)**: `ECO.applyPill` only checks `daoOath` (forgetting `voluntaryVows.VOW_NO_MEDICINE` when personal pills are eaten via `applyQualityPill`!), while `ECO.usePill` checks `voluntaryVows.VOW_NO_MEDICINE`!
- **Exact Fix**:
  - Create a single predicate `CULT.isAbstainingFromPills(n)` checking **both** `(n.daoOath && n.daoOath.upheld &&`

`['VOW_ABSTAIN_PILLS','VEGETARIAN_PACIFIST'].includes(n.daoOath.id))` and `(n.voluntaryVows && n.voluntaryVows.VOW_NO_MEDICINE && n.voluntaryVows.VOW_NO_MEDICINE.active)`.
- Block autonomous pill buying, welfare distribution, hourly feeding, `autoUse`, `personalPillUse`, and `fightUse` whenever `CULT.isAbstainingFromPills(n)` is `true`—routing injured ascetics to **`infirmary` (`A.heal`), `spirit_pond`, or `meditation_hall` (`SERENE_PURIFICATION`)** instead (unless in a lethal `serious: true` fight at `HP < 12%` with `willpower < 45`, where a desperate disciple *chooses* to break their vow to survive!).

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3. `[PILL-REC-03]` Specialized Antidote, Mass-Heal, Meridian-Cure & Curse-Purge Pills (`ITEM190_PILL_001..004, 008`) Flattened Into Weak Solo Generics!

Inspecting `data/data_items.js` vs `js/economy.js:326–352` & `js/cpeco.js:430` reveals **four broken specialty recovery designs**:

2. **Anti-Venom / Anti-Gu Pills Ignored by Poisoned Fighters (`ITEM190_PILL_001` & `ITEM190_PILL_002`)**:
   - `ITEM190_PILL_001` (โอสถหยกขาวชะล้างพิษโลหิต, `supportedToxinTags: ['HEMOTUXIN', 'PILL_TOXICITY']` — note typo `'HEMOTUXIN'` $\to$ `'HEMOTOXIN'`) and `ITEM190_PILL_002` (โอสถบัวหิมะแก้พิษประสาทและกู่, Tier 4, `supportedToxinTags: ['NEUROTOXIN', 'QI_ROT_GU', 'CORPSE_MIASMA']`).
   - **Bug**: `economy.js:329` maps `EFX_MERIDIAN_TOXIN_PURGE` **solely** to `e.clearToxic` (pill toxicity!), **omitting `e.clearPoison` and never clearing `venomSeed`, `crimsonGuSwarm`, `sovereignPlague`, or `ashenToxin`**!
   - As a result, when a disciple is dying of Gu/Venom in combat, `CPE.fightUse` (`cpeco.js:430`: `pickPill(n, 'clearPoison', null)`) **cannot even recognize `ITEM190_PILL_001/002` as antidotes**, and eating them via `applyPill` leaves poison/Gu stacks untouched!
3. **Party Mass-Heal Pill Heals Only 1 Person (`ITEM190_PILL_004` โอสถน้ำค้างทิพย์ฟื้นพลังหมู่, Tier 3)**:
   - Declares `EFX_MASS_VITAL_HEAL` with `{ healAmount: 32, maxRecipients: 4 }`, yet `economy.js:327` converts it to solo `e.healHealth = 32` (`weaker than Tier-1 P:BASIC_HP = 35`!) and **completely ignores `maxRecipients: 4`**!

- **Fix**: When `maxRecipients > 1` is consumed in combat (`f`), expedition, or infirmary, splash `healAmount%` across up to `maxRecipients` injured allies within `4.5` tiles!

4. **Cheap Tier-1 Wound Pill Cures Severe Meridian Ruptures, While Tier-4 Golden Meridian Pill Has No Distinct Hook (`ITEM190_PILL_003` vs `P:BASIC_WOUND`)**:
   - `economy.js:344` (`npc.inj.forEach(i => { i.until -= e.healInjury * 360; })`) lets a cheap Tier-1 `P:BASIC_WOUND` (`healInjury: 30` $\to$ `-7.5 days`) instantly erase **Meridian Overdrive ruptures (ชีพจรปกครองร้าว) and Vow-Shatter spiritual trauma (สัตย์สาบานพังทลาย)**, and forgets to filter out cured injuries (`npc.inj = npc.inj.filter(i => i.until > W.t)`) immediately!
   - **Fix**: Tag internal/spiritual injuries (ชีพจรปกครองร้าว, ธาตุไฟเข้าแทรก, สัตย์สาบานพังทลาย) as `meridianOrSoul: true` (`0.25×` recovery from ordinary Tier-1 wound salves, `1.0×` full instant mend from `curesMeridianWound: true` `ITEM190_PILL_003` / `EFX_INJURY_MEND`), and immediately prune `i.until <= W.t`!
5. **Curse & Elemental Purge Pill Leaves Curses Untouched in `applyPill` (`ITEM190_PILL_008`)**:
   - `economy.js:330` maps `EFX_PURIFY_HOSTILE_STATUS` (`maxDispelTier: 5, maxStatusesRemoved: 3`) to `{ clearHeartDemon: 12, clearPoison: 25 }`, so taking `ITEM190_PILL_008` via `CPE.autoUse`/`fightUse` clears **only `'poison'`**, leaving `'curse'`, `'burn'`, `'bleed'`, `'chill'`, `'stygianFreeze'`, and `'weaken'` active! Remove up to `maxStatusesRemoved` hostile statuses in `ECO.applyPill`!

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4. `[PILL-REC-04]` Nonexistent `'P:HEAL'` Ghost Key in Standing Edicts (`sect.js`), Zero-Qi Samurai Buying `P:BASIC_QI`, & Anti-Heal Bypass!

1. **Ghost Pill Key `'P:HEAL'` in `SECT.ELDER_COUNCIL_EDICTS.executeStandingEdicts` (`js/sect.js`)**:
   - `executeStandingEdicts` checks `P:HEAL < 10` (a nonexistent ID—canonical IDs in `data_cp.js` are **`'P:BASIC_HP'`** and **`'P:BASIC_WOUND'`**!).
   - Because `ECO.count(W, 'P:HEAL')` is `0` on clean worlds (or if minted, `ECO.pillInfo('P:HEAL') === null` makes it **100% inedible and unsellable**!), the Steward wastes monthly edict budget on a ghost key! Replace `'P:HEAL'` with `'P:BASIC_HP'` and `'P:BASIC_WOUND'` (plus add a migration moving any legacy `stock['P:HEAL']` into `stock['P:BASIC_HP']`).
2. **Zero-Qi `SHADOW_SAMURAI` Buys & Hoards `P:BASIC_QI` (`js/cpeco.js:269`)**:

- `CPE.pickPurchase` (`line 269`) evaluates `e.qiRestore && c && have(k) < 1 && n.rank >= 1` without checking `CULT.qiMax(n) > 0`! Every Rank 1+ Bushido Samurai spends CP on `P:BASIC_QI` and carries it unused forever (`0 < 0 * 0.2` never triggers in `fightUse`). Add `CULT.qiMax(n) > 0` guard!

3. **Mid-Combat Pill Healing Ignores `healBlock` / `BLOOD_HALT` / `stygianFreeze` (`economy.js:343`)**:
   - When `ECO.applyPill` heals `s.hp` while `npc.act && npc.act.id === 'fight'`, multiply heal amount by `g.COMBAT && g.COMBAT.healMultiplier ? g.COMBAT.healMultiplier(npc) : 1` so acupoint blood-sealing (`BLOOD_HALT`) and Nether Freeze (`stygianFreeze`) cannot be trivially bypassed by basic pills!

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5. Implementation & Verification Checklist for GPT / Gemini (`06A`)

- [ ] **Fix `qiRestore` Scale & Consolidate `ECO.applyPill` (`economy.js:322, 338`, `cpeco.js:395`)**: Normalize `qiRestore > 1.5 ? qiRestore / 100 : qiRestore` and handle `qiRestore`, `energyRestore`, and `restRestore` directly inside `economy.js`'s `ECO.applyPill`.
- [ ] **Enforce `CULT.isAbstainingFromPills(n)` Across All Autonomous Loops (`sim.js`, `cpeco.js`)**: Prevent `SIM.hourly`, `CPE.welfareDaily`, `CPE.pickPurchase`, `CPE.autoUse`, `A.personalPillUse`, and `CPE.fightUse` from force-feeding pills to vow-holders; unify `daoOath` + `voluntaryVows.VOW_NO_MEDICINE` checks.
- [ ] **Restore Specialty Recovery Pill Mechanics (`ITEM190_PILL_001..004, 008`)**: Clear Gu/venom/miasma statuses on `EFX_MERIDIAN_TOXIN_PURGE` (and set `clearPoison`), splash `maxRecipients: 4` on `ITEM190_PILL_004`, differentiate `curesMeridianWound: true` (`ITEM190_PILL_003`) vs Tier-1 wound salves (and prune expired `npc.inj` immediately), and purge up to `maxStatusesRemoved` debuffs/curses on `ITEM190_PILL_008`.
- [ ] **Fix `'P:HEAL'` Ghost Key (`sect.js`), `SHADOW_SAMURAI` Qi-Pill Filter (`cpeco.js:269`), & Combat `healMultiplier` (`economy.js:343`)**.